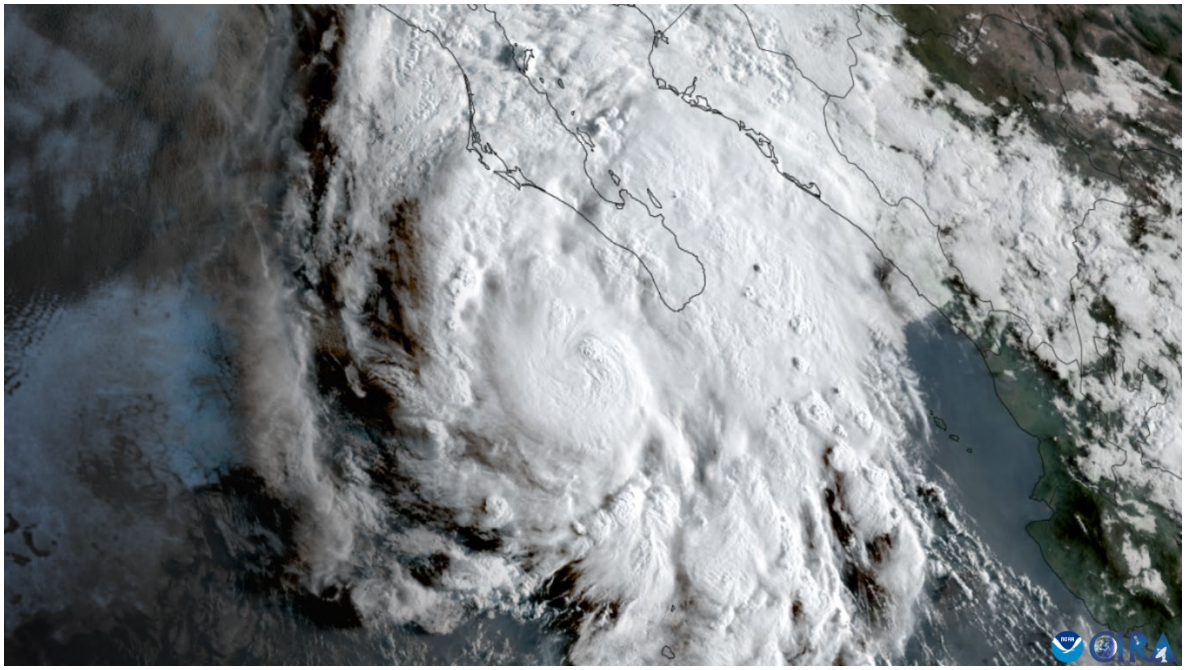
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT

HURRICANE LORENA

(EP122025)

2–4 September 2025

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National Hurricane Center
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GOES-18 GEOCOLOR IMAGE OF HURRICANE LORENA AT 1800 UTC 3 SEPTEMBER 2025. IMAGE COURTESY OF NOAA/CIRA/CSU.

Lorena was a category 1 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that remained just offshore of the Baja California peninsula. While Lorena caused heavy rainfall and flooding over Baja California Sur, no fatalities or injuries were attributable to the cyclone.

Hurricane Lorena

2–4 SEPTEMBER 2025

SYNOPTIC HISTORY

The genesis of Lorena can be traced back to a convectively active tropical wave that emerged off the coast of west Africa on 18 August. The wave continued to produce a large area of disorganized showers and thunderstorms during the next several days while it traveled quickly across the eastern and central tropical Atlantic. The tropical wave came close to becoming a tropical cyclone in the central tropical Atlantic before weakening as it moved across the Windward Islands late on 24 August. Convection became more intermittent as the wave traveled through the Caribbean Sea over the next few days, then became more robust as it crossed Central America beginning late on 26 August. The wave emerged over the eastern Pacific Ocean on 28 August, and convection gradually became better organized during the next several days while the system moved west-northwestward well to the south of Guatemala and southern Mexico. The thunderstorm activity became sufficiently organized by 0000 UTC 2 September for the system to be designated a tropical depression when it was located about 120 n mi south of Manzanillo, Mexico, and it became a tropical storm 6 h later. The "best track" chart of Lorena's path is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[1].

During the next couple of days, Lorena tracked northwestward over very warm waters near 30ºC, within a moist mid-level environment, while influenced by light vertical wind shear. Steady-to-rapid intensification ensued, with the cyclone becoming a category 1 hurricane (on the Saffir-Simpson Hurricane Wind Scale) 24 h after becoming a tropical storm at 0600 UTC 3 September about 85 n mi southwest of Cabo San Lucas, Mexico. The cyclone remained a hurricane for about a day and reached its estimated peak intensity of 75 kt from 1800 UTC 3 September–0000 UTC 4 September. The evolution of a well-defined eyewall feature was captured by radar imagery from Los Cabos, Mexico, on 3 September as shown in Fig. 4. Environmental conditions became hostile rather abruptly on 4 September, as Lorena became influenced by increasingly strong southwesterly vertical wind shear while moving over progressively cooler waters below 26°C. These conditions resulted in rapid weakening of the cyclone, with Lorena losing deep convection shortly after 1800 UTC 4 September and becoming a post-tropical cyclone by 0000 UTC 5 September. The post-tropical cyclone meandered a couple of hundred n mi offshore of the west coast of Baja California for the next day or so, then tracked slowly northwestward and eventually opened into a trough by 1200 UTC 7 September.

[1] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

METEOROLOGICAL STATISTICS

Observations in Lorena (Figs. 2 and 3) include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Observations also include flight-level and dropwindsonde observations from two flights of the 53rd Weather Reconnaissance Squadron of the U.S. Air Force Reserve Command (AFRES) (Fig. 5). Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Lorena.

The container ship **OOCL Egypt** (call sign **VRLZ3**) reported a sustained tropical-storm-force wind of 37 kt in association with Lorena at 1200 UTC 2 September while located about 85 n mi west of Manzanillo, Mexico. There were no land-based reports of sustained tropical-storm-force winds associated with Lorena. Tropical-storm-force gusts were reported across the southern portion of Baja California Sur, however, with a 53-kt wind gust measured in Cabo San Lucas and a 44-kt gust measured in Sierra La Laguna (Table 2).

Winds and Pressure

Lorena's estimated peak intensity of 75 kt from 1800 UTC 3 September to 0000 UTC 4 September is based on a blend of aircraft data and TAFB subjective Dvorak intensity estimates. The two reconnaissance missions by the AFRES into Lorena provided valuable data. The first mission sampled peak flight-level winds of 79 kt, 6 n mi to the northeast of the flight-level center at 1700 UTC 3 September. Using standard reduction factors from flight-level (700 mb), the surface winds were estimated to be around 71 kt. During this same mission, a dropsonde measured a 70-kt surface wind in the northeast eyewall. The WL150 and MBL winds from this dropsonde both suggested surface winds of 64 and 63 kt respectively. These data support a peak intensity of 75 kt when accounting for undersampling, and are further supported by the TAFB subjective Dvorak intensity estimates of T4.5/77 kt at 1800 UTC 3 September and 0000 UTC 4 September.

Lorena's minimum pressure of 981 mb was based on the Knaff-Zehr-Courtney pressure-wind relationship. This pressure occurred at 0000 UTC 4 September, late in the afternoon when the diurnal pressure cycle is at its lowest point. The first AFRES mission into Lorena measured a minimum central pressure of 985 mb with 7 kt of wind at the surface (10 m) at 1759 UTC 3 September, which supports a minimum central pressure of 984 mb at that time.

Rainfall and Flooding

Heavy rainfall affected portions of southern Baja California Sur (Fig. 6) as Lorena passed about 100 n mi to the west. The highest reported storm-total rainfall was 12.28 inches (312.0 mm) at Santa Anita, followed by 10.43 inches (265.0 mm) at Yeneka and 10.24 inches (260.0 mm) at Boca del Salado. Additional notable totals included 8.98 inches (228.0 mm) at San José del Cabo and 7.86 inches (199.6 mm) at El Sauzal (Table 2). These very heavy rains produced flooding across portions of the municipality of Los Cabos and surrounding areas.

CASUALTY AND DAMAGE STATISTICS

No fatalities or injuries can confidently be attributed to Lorena. There were several media reports indicating Lorena was responsible for several fatalities in the Mexican states of Nayarit and Sinaloa. However, after reviewing the articles and cross-referencing satellite imagery for the dates and times when the fatalities occurred, it was determined that these deaths were neither directly or indirectly related to Lorena. In Baja California Sur, impacts were concentrated in the municipalities of Los Cabos, La Paz, Mulegé, Loreto, and Comondú. Heavy rainfall and flooding resulted in water inundation of at least eight homes, while three additional residences were reported to have been swept away by floodwaters. Civil protection authorities reported approximately 50 vehicles stranded due to flooded or damaged roadways. Four landslides were also documented, and there was a minor disruption to electrical service associated with damage to a single power pole. Roadways across Los Cabos were engulfed in thick mud and debris due to the flooding, necessitating a substantial response and recovery effort. A total of 119 people sought refuge in four temporary shelters during the event. Numerous federal, state, and municipal agencies participated in preparedness and emergency response activities.

FORECAST AND WARNING CRITIQUE

The genesis of Lorena was well anticipated. The tropical wave from which Lorena developed was first highlighted in the Tropical Weather Outlook (TWO) 156 h prior to formation (Table 3). The 7-day genesis probability was raised to the medium category 114 h before formation and to the high category 72 h in advance. Two-day genesis forecasts were also excellent. Low probabilities were introduced 78 h prior to formation, medium probabilities 48 h in advance, though high probabilities were introduced only 6 h before development occurred.

Forecasts of Lorena's genesis location were highly accurate, with the system forming within each of the genesis forecast areas depicted in NHC's Graphical TWO, resulting in a 100% hit rate (Fig. 7).

A verification of NHC official track forecasts for Lorena is given in Table 4a. Official track forecast errors were higher than the mean official errors for the previous 5-yr period at all forecast intervals. This is likely attributable to a bias in some of the model guidance keeping Lorena a

deeper cyclone longer and steering the system into Baja California, while in actuality Lorena weakened much faster than anticipated and remained well offshore of Baja California steered by the low-level flow. A homogeneous comparison of the official track errors with selected guidance models is given in Table 4b. The official track forecasts outperformed all of the single-model aids at all forecast intervals, with the exception of EMXI and HWFI which performed better than the official forecast for a couple of periods. The consensus models generally outperformed the official track forecasts. The Google DeepMind Ensemble Mean (GDMI) was particularly skillful, outperforming the official forecasts at all forecast intervals, and also correctly predicted the cyclone would stall off the coast of Baja California. The Hurricane Forecast Improvement Program (HFIP) Corrected Consensus Approach (HCCA), Florida State Super Ensemble (FSSE), and GFEX models also performed better than the official forecasts at most time intervals.

A verification of NHC official intensity forecasts for Lorena is given in Table 5a. Official intensity forecast errors were higher than the mean official errors for the previous 5-yr period at all forecast intervals, except for the 60-h interval. The larger than normal forecast errors are likely attributable to a bias in some of the models keeping the system stronger for a longer period than what actually occurred. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 5b. The official intensity forecasts were outperformed by a considerable number of guidance models, with the HCCA and FSSE models performing better than the official forecast at all forecast intervals, and the ICON, IVCN, and IVDR models outperforming the official forecast at most time intervals.

Watches and warnings associated with Lorena are given in Table 6.

ACKNOWLEDGEMENTS

Radar imagery shown in Fig. 4 was provided by CONAGUA, the National Meteorological Service of Mexico and Brian McNoldy. Dr. Lisa Bucci (NHC) created the recon summary graphic (Fig. 5). Rainfall data shown in Fig. 6 were provided by CONAGUA. Dr. Philippe Papin (NHC) created the GTWO verification graphic (Fig. 7).

Table 1. Best track for Hurricane Lorena, 2–4 September 2025

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
02 / 0000	16.9	104.8	1005	30	tropical depression
02 / 0600	17.3	106.3	1004	35	tropical storm
02 / 1200	18.0	107.4	1003	40	"
02 / 1800	18.9	108.3	1001	45	"
03 / 0000	19.8	109.2	998	55	"
03 / 0600	20.8	110.0	991	65	hurricane
03 / 1200	21.9	111.1	987	70	"
03 / 1800	22.5	112.1	984	75	"
04 / 0000	23.1	112.8	981	75	"
04 / 0600	23.6	113.5	984	70	"
04 / 1200	24.0	114.0	990	60	tropical storm
04 / 1800	24.4	114.6	996	45	"
05 / 0000	24.5	115.0	998	40	low
05 / 0600	24.4	115.0	1001	35	"
05 / 1200	24.3	114.9	1002	30	"
05 / 1800	24.4	114.5	1005	30	"
06 / 0000	24.7	114.7	1007	25	"
06 / 0600	25.0	115.1	1007	25	"
06 / 1200	25.2	115.5	1007	20	"
06 / 1800	25.5	115.9	1008	20	"
07 / 0000	25.8	116.5	1009	15	"
07 / 0600	26.1	117.0	1009	15	"
07 / 1200					dissipated

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
04 / 0000	23.1	112.8	981	75	maximum winds and minimum pressure

Table 2. Selected surface observations for Hurricane Lorena, 2–4 September 2025.

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)	
Mexico						
Baja California Sur						
Santa Anita						12.28
Yeneka						10.43
Boca del Salado						10.24
San Jose del Cabo						8.98
El Sauzal						7.86
Cabo San Lucas			04/0210		53	
Sierra La Laguna			04/0740		44	

[a] Date/time is for sustained wind when both sustained and gust are listed.

Table 3. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	78	156
Medium (40%-60%)	48	114
High (>60%)	6	72

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Lorena, 2–4 September 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	23.7	43.0	58.1	66.6	73.3			
OCD5	42.0	88.7	125.2	188.4	215.3			
Forecasts	10	8	6	4	2			
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 4b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Lorena, 2–4 September 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	23.3	46.7	53.9	54.2				
OCD5	41.9	84.2	102.3	170.3				
GFSI	30.1	64.8	73.3	98.8				
HMNI	27.3	48.7	58.9	67.8				
HFAI	29.4	66.0	93.0	120.5				
HFBI	23.5	59.4	87.8	83.8				
HWFI	27.4	58.6	**45.3**	**34.2**				
EMXI	**22.1**	51.9	76.4	96.9				
CMCI	39.1	66.5	90.8	102.3				
AEMI	28.2	54.9	64.6	71.8				
HCCA	**19.6**	**38.2**	**51.5**	61.2				
FSSE	**21.0**	**43.6**	**43.5**	56.0				
GFEX	**20.5**	**39.6**	**43.5**	55.3				
TVCE	**21.4**	47.0	55.8	63.4				
TVCX	**20.9**	**45.9**	55.5	62.6				
TVDG	**19.2**	**43.9**	**52.9**	62.3				
TABD	38.9	81.6	103.4	125.6				
TABM	27.2	47.2	62.6	91.9				
TABS	25.4	**44.4**	67.4	108.7				
GDMI	**19.0**	**36.1**	**53.0**	**47.4**				
Forecasts	8	6	4	2	0	0	0	0

Table 5a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Lorena, 2–4 September 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	6.5	14.4	18.3	15.0	**7.5**			
OCD5	10.3	15.4	18.5	18.0	6.5			
Forecasts	10	8	6	4	2			
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 5b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Lorena, 2–4 September 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 5a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	6.9	15.0	18.8	17.5				
OCD5	10.9	**13.8**	**14.0**	**9.5**				
HWFI	10.0	**9.3**	**5.8**	**4.5**				
HMNI	12.8	19.2	**14.8**	**3.5**				
HFAI	9.4	15.7	**17.2**	**13.5**				
HFBI	8.6	**11.7**	**12.5**	20.0				
DSHP	7.6	**12.0**	20.8	28.0				
LGEM	8.2	**12.0**	20.0	18.0				
ICON	8.0	**8.7**	**12.0**	**12.5**				
IVCN	8.1	**9.0**	**7.2**	**12.5**				
IVDR	8.8	**10.5**	**7.5**	**10.5**				
GFSI	12.2	15.0	**12.2**	**8.5**				
EMXI	14.6	21.5	21.5	17.5				
HCCA	**5.1**	**4.5**	**8.5**	**16.5**				
FSSE	**6.4**	**9.0**	**8.0**	**8.5**				
GDMI	13.5	18.7	**16.5**	**8.0**				
Forecasts	8	6	4	2				

Table 6. Watch and warning summary for Hurricane Lorena, 2–4 September 2025.

Date/Time (UTC)	Action	Location
2 / 2100	Tropical Storm Watch issued	Cabo San Lucas to Cabo San Lazaro
3 / 0900	Tropical Storm Watch modified to	Cabo San Lucas to Santa Fe
3 / 0900	Tropical Storm Warning issued	Santa Fe to Cabo San Lazaro
3 / 0900	Tropical Storm Watch issued	Cabo San Lazaro to Punta Abreojos
3 / 1500	Tropical Storm Watch modified to	Punta Abreojos to Punta Eugenia
3 / 1500	Tropical Storm Warning modified to	Santa Fe to Punta Abreojos
3 / 2100	Tropical Storm Watch modified to	Todos Santos to Santa Fe
3 / 2100	Tropical Storm Watch discontinued	Punta Abreojos to Punta Eugenia
3 / 2100	Tropical Storm Watch issued	Bahia De Los Angeles to Santa Rosalia
3 / 2100	Tropical Storm Warning modified to	Santa Fe to Punta Eugenia
4 / 0600	Tropical Storm Watch discontinued	Todos Santos to Santa Fe
4 / 1500	Tropical Storm Watch discontinued	All
4 / 1500	Tropical Storm Warning modified to	Puerto Cortes to Punta Eugenia
4 / 2100	Tropical Storm Warning discontinued	All

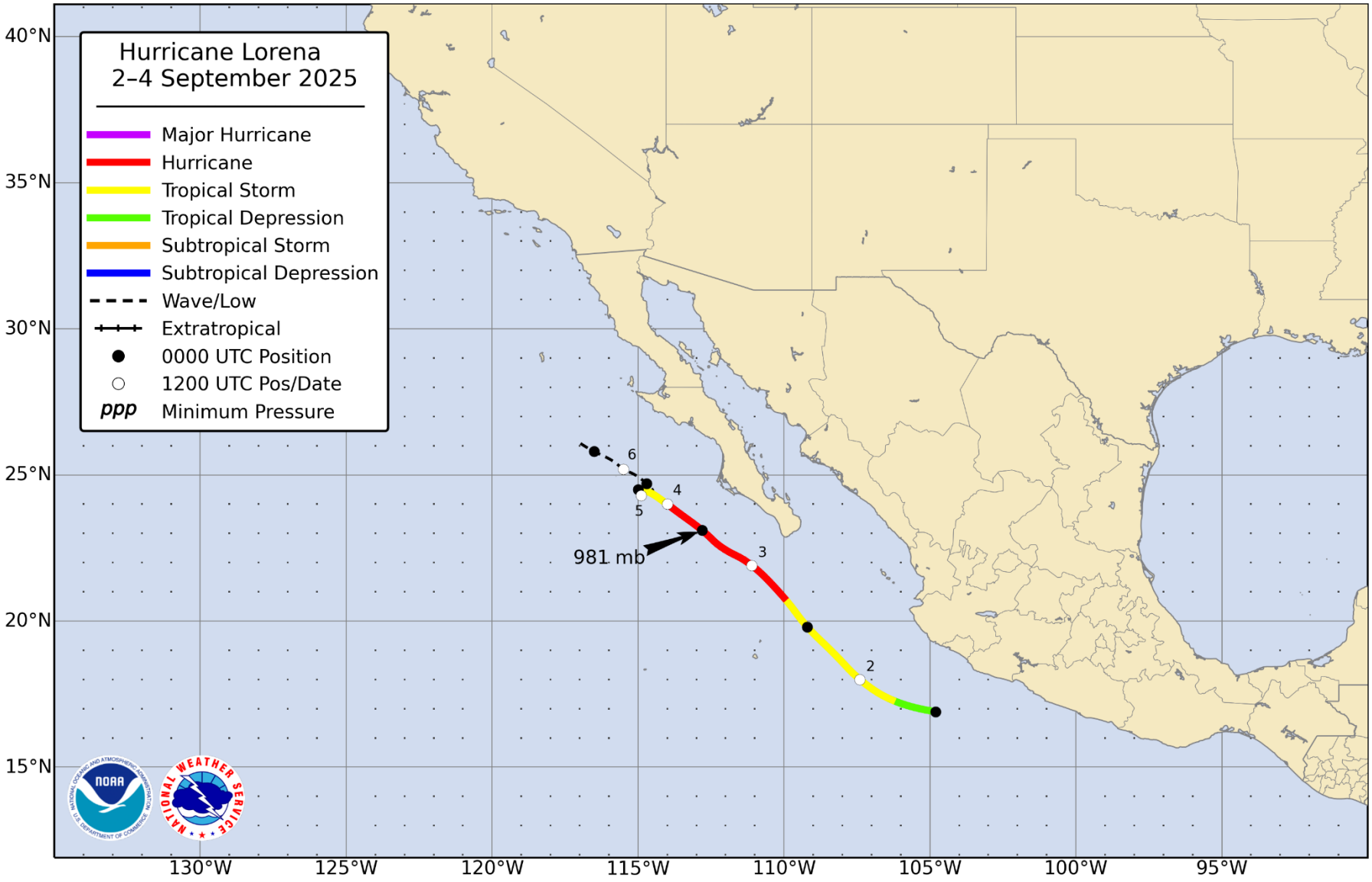


Figure 1. Best track positions for Hurricane Lorena, 2–4 September 2025.

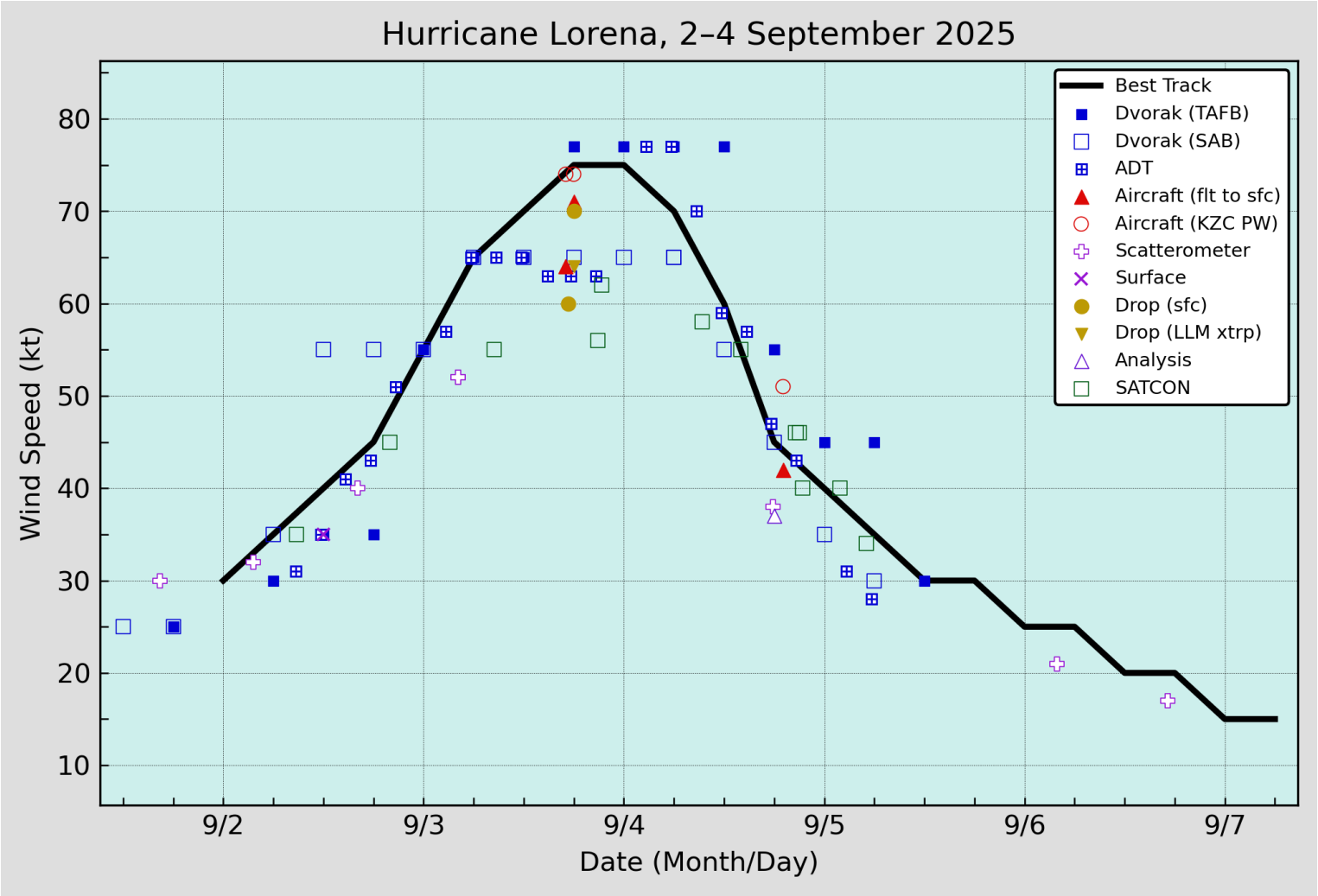


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve Hurricane Lorena, 2–4 September 2025. Aircraft observations have been adjusted for elevation using a 90% adjustment factor for observations from 700 mb. Dropwindsonde observations include actual 10 m winds (sfc), as well as surface estimates derived from the mean wind over the lowest 150 m of the wind sounding (LLM). Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

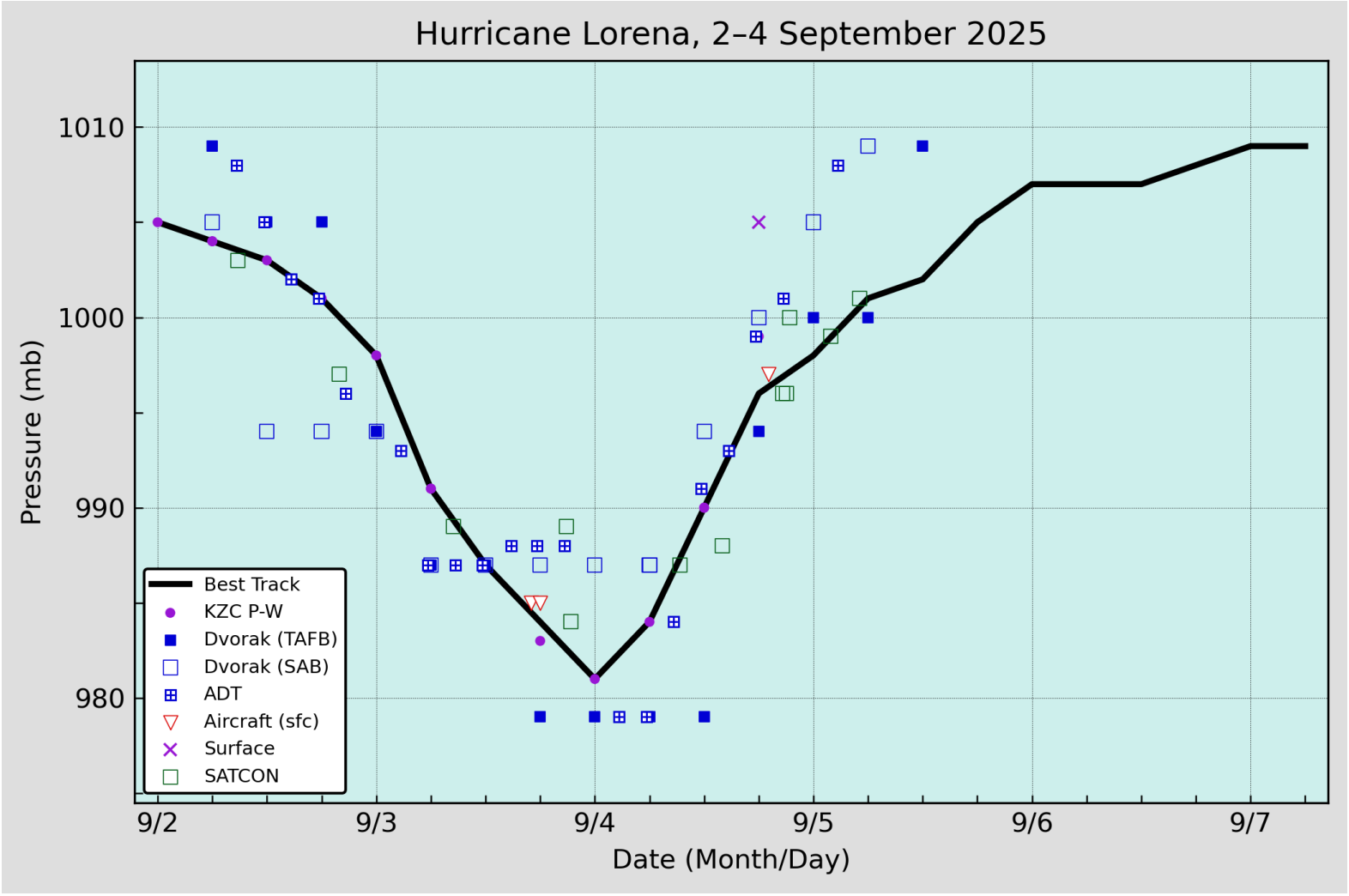


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Lorena, 2–4 September 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

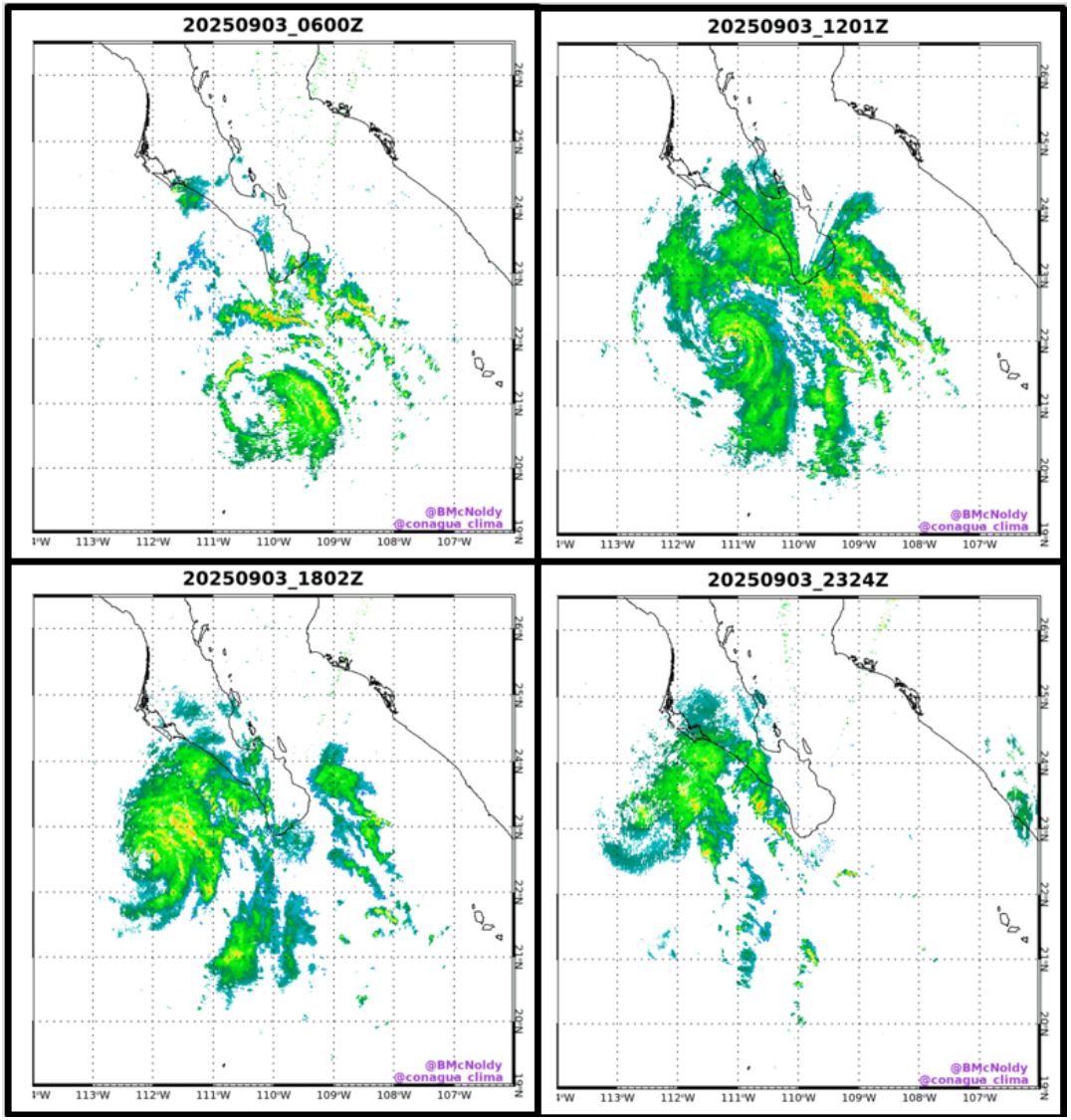


Figure 4. Radar images from Los Cabos, Mexico, of Hurricane Lorena around peak intensity on 3 September 2025. Image courtesy of CONAGUA, the National Meteorological Service of Mexico and B. McNoldy.

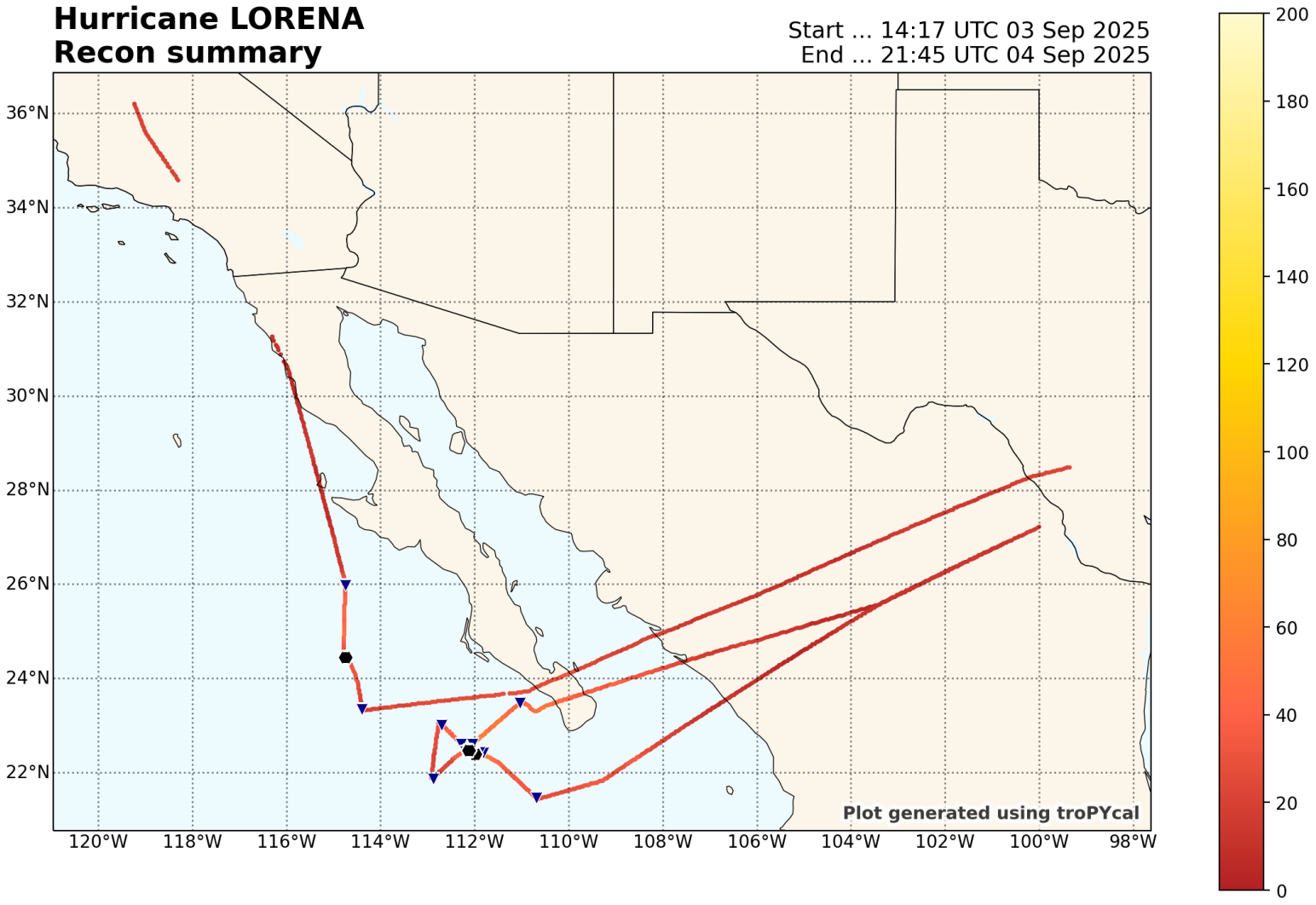


Figure 5. Air Force Reserve aircraft flight tracks (colored lines) from reconnaissance missions into Hurricane Lorena from 3 and 4 September 2025. The blue triangles indicate dropsonde locations. The color of the flight track represents the observed flight-level wind speed in knots at that location (see legend).

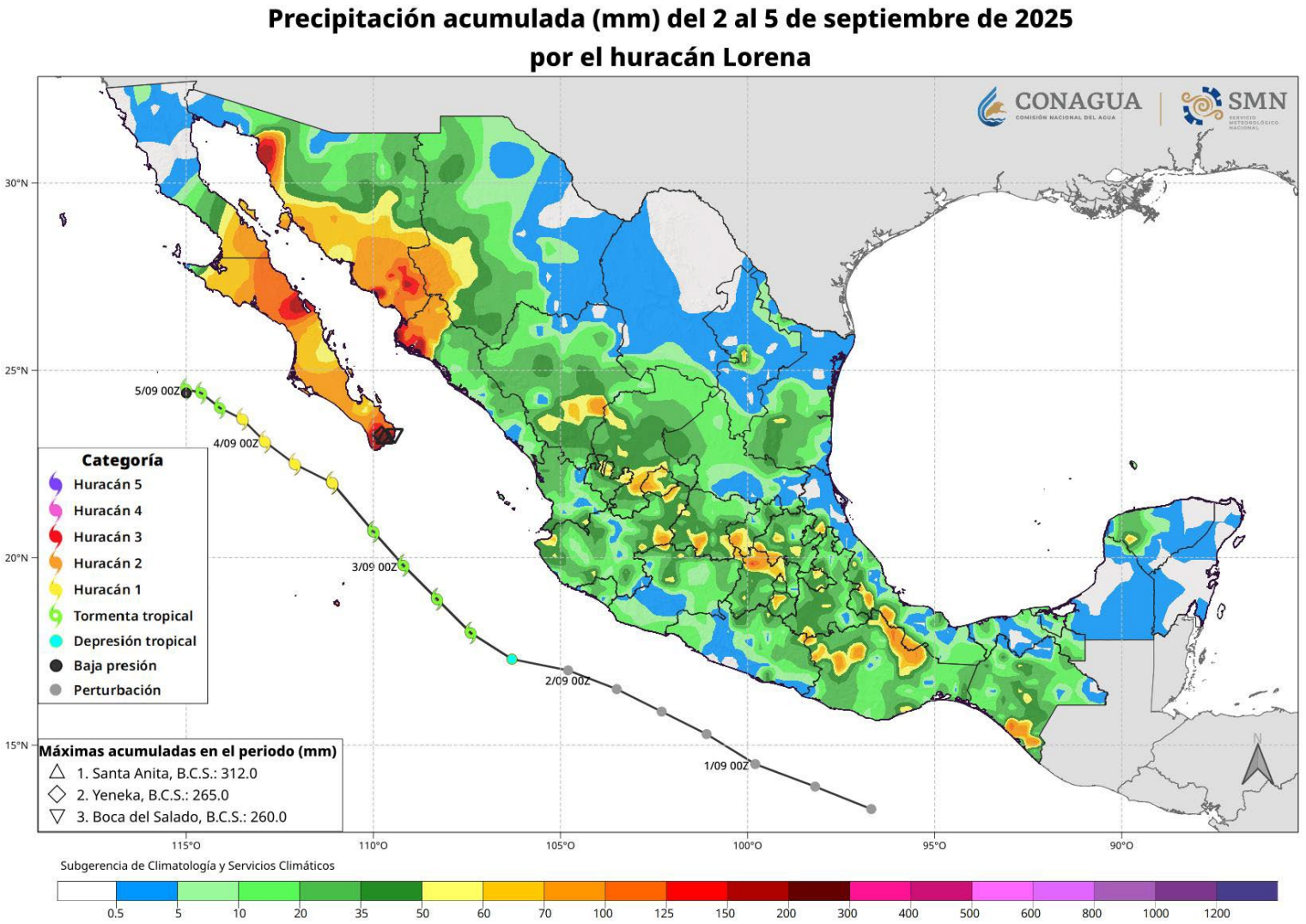


Figure 6. Rainfall accumulations (mm) in Mexico from 2–5 September 2025. Track and intensity are based on the operational NHC assessment. Not all of the rainfall depicted here is directly related to Lorena. Image courtesy of CONAGUA, the National Meteorological Service of Mexico.

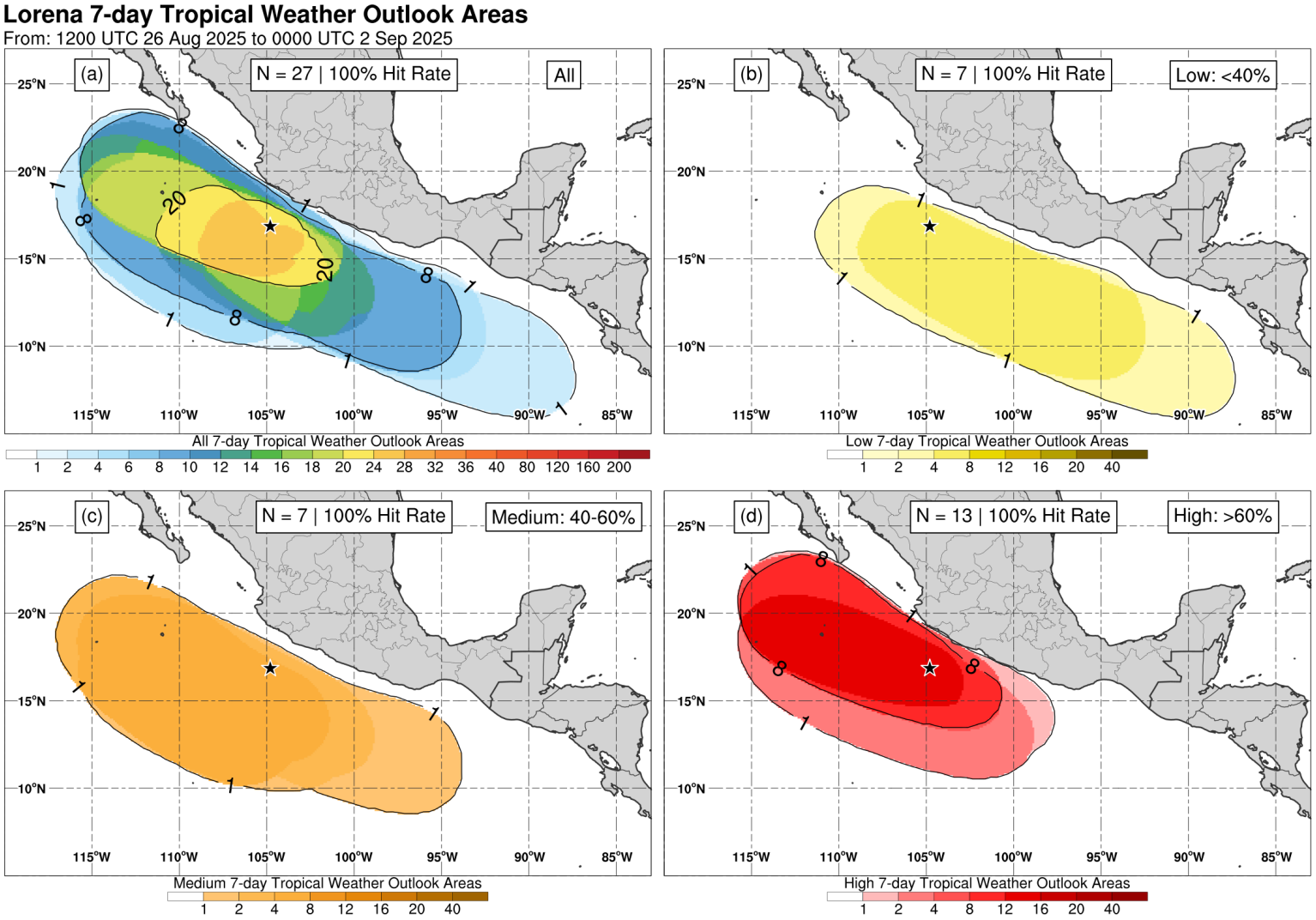


Figure 7. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Hurricane Lorena for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.